

Zein Hajj-Ali

☎ (514)-814-4665 | ✉ zeinhajjali@outlook.com | [in linkedin.com/in/zeinhajjali](https://www.linkedin.com/in/zeinhajjali) | 🌐 zeinh.ca

PROFESSIONAL SUMMARY

Senior AI/ML Engineer & Consultant with hands-on experience delivering production ML systems and building platform services (pipelines, deployment, monitoring) that make model integration reliable and robust. Strong Python and systems design foundation, comfortable with ambiguity, and experienced partnering with stakeholders to take AI capabilities from prototype to production.

WORK EXPERIENCE

Senior Developer & Consultant, AI/ML May 2025 — Feb 2026

Artificial Intelligence Developer & Consultant Apr 2023 — May 2025

 Lemay.ai *Ottawa, ON*

- Joined an established graph-based RAG platform and took ownership of new capabilities, including LLM inference workflows (vLLM) and knowledge graph population pipelines built on Neo4j.
- Designed and implemented document extraction and parsing pipelines, transforming unstructured documents into structured entities and relationships that extended and enriched the existing knowledge graph.
- Improved retrieval and reasoning quality by integrating structured graph data into downstream LLM workflows, supporting more precise and context-aware responses.
- Took ownership of an existing LLM-driven answer retrieval system for a large client, building new features and productionizing it to deliver grounded, exact responses in high-precision use cases.
- Implemented multi-stage LLM prompting pipelines to reformulate and expand user queries, feeding both semantic search and exact matching systems prior to result ranking.
- Deployed and operated ML and LLM services using Python, Docker, Airflow, and MLflow, supporting reproducible experiments and stable production deployments.
- Worked closely with stakeholders to scope features, communicate system behavior, and evolve solutions based on real-world usage and constraints.

Contract Research Assistant Jan 2023 — Apr 2023

 Carleton University - National Research Council Canada *Ottawa, ON*

- Began research on building a data collection pipeline, allowing for scaling up smaller Google Street View datasets.
- Contributed to the creation of a novel dataset for computer vision tasks, focusing on Google Street View imagery.
- Developed and applied annotation protocols for labeling images with regions-of-interest, ensuring high-quality data for future ML model training.

Research Assistant Sep 2020 — Dec 2022

 Carleton University Biomedical Informatics Collaboratory *Ottawa, ON*

- Conducted impactful research on non-contact patient monitoring using depth cameras, resulting in multiple peer-reviewed publications.
- Developed and deployed state-of-the-art Vision Transformer (ViT) and Convolutional Neural Network (CNN) models for accurate depth data prediction.
- Developed novel depth data transformation method, enabling effective use of non-uniform data for precise vital sign estimation.
- Utilized a modified Scrum/Sprint methodology to consistently achieve research goals and deadlines.

Teaching Assistant Sep 2021 — May 2022

 Carleton University *Ottawa, ON*

- Collaborated effectively with instructors to prepare and update comprehensive course materials and lesson plans.
- Delivered clear and concise lab instructions, providing essential technical support to students.

- Provided thorough grading and constructive feedback across diverse assignments and modalities, contributing to student understanding and improvement.

Junior Software Developer

Apr 2020 — Aug 2020

 Spreveel Inc.

Ottawa, ON

- Contributed directly to the development of client-facing web and mobile applications, enhancing user experience and functionality.
- Maintained and significantly improved the functionality of machine-learning pipelines in development, contributing to overall system performance and stability.

EDUCATION

Carleton University

Ottawa, ON

M.A.Sc. in Electrical and Computer Engineering

2022

- Data Science Specialization from the Carleton University Institute for Data Science (CUIDS)
- Awarded: Queen Elizabeth II - Graduate Scholarship in Science and Technology (QEII-GSST)
- Supervisor: James R. Green, PhD, P.Eng., SMIEEE

Carleton University

Ottawa, ON

B.Eng. in Computer Systems Engineering

2019

SKILLS & ABILITIES

- *LLMs & Retrieval Systems*: vLLM, retrieval-augmented generation (RAG), semantic search, prompt orchestration.
- *Machine Learning*: PyTorch, TensorFlow, Keras, SciKit-Learn, deep learning, computer vision, reinforcement learning, multi-GPU and distributed training.
- *Cloud & MLOps*: Docker / containerization, Kubernetes, virtual machine management, MLflow, AirFlow.
- *Data Processing*: NumPy, Pandas, SQL, data pipeline development and automation.
- *Programming Languages*: Python (primary), JavaScript; working knowledge of Java and C++.
- *Version Control & CI/CD*: Git, DVC, experience with automated testing and deployment workflows.
- *Frontend & Prototyping*: Gradio, React, React Native/Expo.
- *Product & Stakeholder Collaboration*: Owned AI initiatives end-to-end, partnered with clients and internal stakeholders, and communicated complex system behavior to technical and non-technical audiences.

PUBLICATIONS

- [1] Z. Hajj-Ali, "Depth-Based Patient Monitoring in the NICU with Non-Ideal Camera Placement," Master of Applied Science, Ottawa, Ontario, 2023. doi: [10.22215/etd/2023-15371](https://doi.org/10.22215/etd/2023-15371).
- [2] Z. Hajj-Ali, K. Greenwood, J. Harrold, and J. R. Green, "Towards Depth-based Respiratory Rate Estimation with Arbitrary Camera Placement," in *2022 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*, June 2022, pp. 1–6. doi: [10.1109/MeMeA54994.2022.9856449](https://doi.org/10.1109/MeMeA54994.2022.9856449).
- [3] Z. Hajj-Ali, D. Kyrollos, and Y. Souley Dosso, "Multi-Modal Registration Techniques for Non-Contact Neonatal Patient Monitoring." CASCONxEVOKE 2021, Nov. 2021.
- [4] Z. Hajj-Ali and G. Wainer, "Spatial Models and Masks in Indoor Analysis for the Spread of COVID-19," in *2021 Winter Simulation Conference (WSC)*, Dec. 2021, pp. 1–12. doi: [10.1109/WSC52266.2021.9715334](https://doi.org/10.1109/WSC52266.2021.9715334).

SELECTED OPEN-SOURCE PROJECTS

Main Contributor/Researcher, DepthSliceTool (github.com/ZeinHajjAli/DepthSliceTool)

- Tooling for depth-data processing and analysis used in computer vision research.

Main Contributor, SlackNotifPy (github.com/ZeinHajjAli/SlackNotifPy)

- Python package for sending customizable Slack notifications from scripts and applications.

PROFESSIONAL DEVELOPMENT

ISO 13485: Quality Management System Training

2024